

MUKESH KASIMAHANATHI

Full Stack and AI Platform Engineer

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SUMMARY

Full-stack engineer with 4 years of production experience across Angular, React, React Native, Node.js, distributed queues, Temporal workflows, AI agent infrastructure, and multi-environment cloud deployments. Comfortable owning systems end to end: frontend, backend, data, AI execution, and release.

EXPERIENCE

AI Platform Engineer (2024 - Present)

- Distributed agent execution platform on Temporal, Kafka, SQS, Redis, and KeyDB.
- LLM rate-limit control across OpenAI, Anthropic, Gemini, and Amazon Bedrock.
- KEDA-driven queue-aware autoscaling and DLQ replay flows for failed jobs.
- Payload codec that offloads large prompts and workflow state to S3 transparently.

Full-Stack Engineer (2023 - 2024)

- Apartment operations platform: Angular + DevUI / Nebular + Node / Express + Sequelize.
- Dynamic FormArray flows, water bill and maintenance calculation, Excel reporting.
- Project Checklist Review System with RBAC, L1/L2/L3 review stages, React + MUI.

Frontend / Mobile Engineer (2022 - 2023)

- React Native cross-platform mobile flows with typed navigation and forms.
- Angular admin tooling and TypeScript fixes across product surfaces.

SELECTED PROJECTS

AI Agent Workflow Platform - Temporal, Kafka, SQS, Redis, KeyDB, KEDA, Bedrock

Apartment Maintenance Management System - Angular, Node.js, Sequelize, Excel export

Project Checklist Review System - React, Node.js, MongoDB, MUI, RBAC

Cloud Deployment Automation - Docker, GitHub Actions, ECR, AWS, SSL

Unified Social Media Hub - Node.js, OAuth, feed and messaging architecture

TECH STACK

Frontend: Angular, React, React Native, TypeScript, Tailwind, Material UI, DevUI, Nebular

Backend: Node.js, Express.js, Sequelize, REST APIs, RBAC, OAuth, JWT

Data: PostgreSQL, Amazon Aurora, MongoDB, Redis, KeyDB

Messaging: Temporal, Kafka, Amazon SQS, DLQ design, idempotency, payload codecs

AI: OpenAI, Claude, Gemini, Amazon Bedrock, token buckets, fallback routing

Cloud: AWS, Docker, Kubernetes, KEDA, Cloud Run, ECR, GitHub Actions, NGINX, SSL

ARCHITECT MINDSET

- Idempotency by default: replay-safe consumers, dedup keys, deterministic workflows.
- Fail loud, isolate fast: jittered retries, DLQ routing, structured failure context.
- Provider-agnostic AI: prompts and tools live above providers; fallback is a config change.
- Observable from day one: queue depth, P95, error rate, and token cost streamed pre-launch.